

1. Annotations define metadata in Spring to configure components, dependency injection, security rules, and request mappings without XML.
2. TLS encrypts communication. PKI manages keys and certificates. A certificate binds identity to a public key. Public keys encrypt/verify, private keys decrypt/sign. Signatures ensure integrity and authenticity.
3. HTTPS was enabled by generating a self-signed certificate, packaging it as a JKS keystore, configuring Spring Security to use it, and trusting the certificate locally so TLS verification succeeds.
4. Common authentication/authorization status codes include 401 Unauthorized and 403 Forbidden.
5. Authentication verifies who you are; authorization checks what you can access. Key components include AuthenticationManager, AuthenticationProvider, UserDetailsService, and SecurityFilterChain.
6. HTTP Session stores user state on the server across multiple requests using a session ID.
7. Cookies are small key-value pairs stored on the client and sent with each HTTP request.
8. Sessions store data server-side and are more secure; cookies store data client-side and are size-limited.
9. Google SSO works via OAuth 2.0: the site redirects to Google, user authenticates, Google returns an authorization code, and the site exchanges it for tokens.
10. Session IDs or secure cookies are used to identify users and retrieve stored user data across requests.
11. Spring Security filters intercept requests to perform authentication, authorization, and security checks.
12. A bearer token is an access token sent in headers. JWT is a signed token containing claims, verified without server-side state.
13. Passwords are hashed with salts (e.g., BCrypt). Credit cards are encrypted or tokenized, never stored in plain text.
14. UserDetailsService loads user data, AuthenticationProvider authenticates, AuthenticationManager coordinates, and filters trigger authentication.
15. Sessions consume server memory and are hard to scale; stateless JWT or distributed session stores solve this.
16. Values are injected from application.properties using @Value or Environment.
17. HttpSecurity defines web security rules; AuthenticationManagerBuilder configures authentication sources.
18. Spring Security is cross-cutting, handling authentication, authorization, and protection across layers.
19. Secrets should be stored in vaults or environment variables, encrypted at rest, and never hard-coded.

Screenshots for HTTPS and Google SSO:

▼ General	
Request URL	https://accounts.google.com/o/oauth2/auth?client_id=975508891494.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Ffleetcode.com%2Faccounts%2Fgoogle%2Flogin%2Fcallback%2F&scope=email&response_type=code&state=CKAU1W4ytYHP
Request Method	GET
Status Code	● 302 Found
Remote Address	[2607:f8b0:4002:c10::54]:443
Referrer Policy	strict-origin-when-cross-origin
▼ Response Headers	
Alt-Svc	h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
Cache-Control	no-cache, no-store, max-age=0, must-revalidate
Content-Encoding	gzip
Content-Length	887
Content-Security-Policy	script-src 'report-sample' 'nonce-LIGSanon6QpNd6xSJm4sBw' 'unsafe-inline' 'unsafe-eval';object-src 'none';base-uri 'self';report-uri /cspreport
Content-Security-Policy	require-trusted-types-for 'script';report-uri /cspreport
Content-Type	text/html; charset=UTF-8
Cross-Origin-Opener-Policy-Report-Only	same-origin; report-to="coop_gse_qebhlk"
Date	Mon, 02 Feb 2026 19:39:16 GMT
Expires	Mon, 01 Jan 1990 00:00:00 GMT
Location	https://accounts.google.com/v3/signin/accountchooser?client_id=975508891494.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Ffleetcode.com%2Faccounts%2Fgoogle%2Flogin%2Fcallback%2F&scope=email&response_type=code&state=CKAU1W4ytYHP

X Headers Preview Response Initiator Timing Cookies	
▼ General	
Request URL	https://www.zoom.us/google_oauth_signin
Request Method	GET
Status Code	● 302 Found
Remote Address	[2407:30c0:182::aa72:3402]:443
Referrer Policy	origin-when-cross-origin
Request URL	https://www.zoom.us/google_oauth_signin
▼ Response Headers	
Alt-Svc	h3=":443"; ma=86400
Cf-Cache-Status	DYNAMIC
Cf-Ray	9c7c2f52f5f116a8-SJC
Content-Length	0
Content-Security-Policy	upgrade-insecure-requests; default-src https://*.zoom.us https://zoom.us blob: 'self'; img-src https: about: blob: data: 'self'; style-src https: safari-extension: chrome-extension: 'unsafe-inline' data: 'self'; font-src https: safari-extension: chrome-extension: blob: data: 'self'; connect-src * about: blob: data: 'self'; media-src * rtmp: blob: data: 'self'; frame-src https: ms-appx-web: zoommtg: zoomus: wvjbscheme: zoomprc: zoomzccprc: data: blob: 'self'; object-src 'none'; base-uri 'none';
Date	Mon, 02 Feb 2026 19:44:44 GMT
Location	https://accounts.google.com/o/oauth2/v2/auth?response_type=code&access_type=offline&client_id=849883241272-ed6lnodi1grnoomiuknqkq2rbvd2udku.apps.googleusercontent.com&prompt=select_account&scope=profile%20email&redirect_uri=https%3A%2F%2Fzoom.us%2Fgoogle%2Foauth&state=WXd2YmtlWHIRYkd3Q0RjLUNVOUVtUSxnb29nbGVfc2lnbmhlu&_x_zm_rtaid=sC6P9TG1SOKxAulvc2plA.1770061475932.63dda1d5c7465cf86d52b3d6fd70feec&_x_zm_rht

HTTP

https://localhost:8443/hello

Save

Share

GET

https://localhost:8443/hello

Send

Docs

Params

Auth

Headers (6)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Descrip...	...	Bulk Edit
	Key	Value	Description		

Body

200 OK

98 ms

358 B

...

Raw

Preview

Visualize

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